Development and Validation of a Inspection Tool for 5G Core Network Authentication Bypass*

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Abstract

Private 5G networks are increasingly used in industrial environments such as smart factories, autonomous systems, and IoT-based services. Since the 5GC(5G Core Network) manages UE registration, authentication, security mode establishment, and session management, it is important to verify whether an abnormal UE can access the network without completing authentication. This paper proposes a UERANSIM-based inspection tool for testing authentication bypass attempts in a 5G core network. The proposed tool modifies the NAS message processing behavior of the UE simulator so that it sends a Security Mode Complete message without transmitting a normal Authentication Response. The experiment was conducted in an Open5GS-based 5G testbed, and the results were analyzed using Wireshark packet capture and core log output. Experimental results show that the Open5GS core network did not send Registration Accept or proceed to PDU Session Establishment for the abnormal UE, confirming that the network blocked access from a UE that had not completed authentication.

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1 Introduction

5G networks provide ultra-high speed, ultra-low latency, and massive connectivity, enabling various industrial services such as smart factories, autonomous vehicles, and IoT(Internet of things) based automation[1]. In particular, private 5G networks[2] can be deployed in limited areas and customized for specific industrial requirements. Because these networks are often used for critical services, reliable authentication and access control are essential.

The 5G Core is responsible for key control functions such as UE(User Equipment) registration, subscriber authentication, security context establishment, and session management. Among them, authentication determines whether a UE is allowed to access the network. If a UE proceeds to later procedures without completing authentication, unauthorized access or security policy bypass may occur. If this requirement is not properly enforced, an attacker may impersonate a legitimate UE and obtain unauthorized network access. This can lead to more severe vulnerabilities, such as unauthorized use of private 5G services, abuse of network resources, unintended charging, and potential session establishment without proper subscriber verification.

Therefore, this paper proposes a security inspection tool that reproduces an authentication bypass attempt and verifies the response of the 5G core network in an open-source testbed environment.

2 Inspection tool design

The proposed inspection tool consists of a UE simulator, a gNB simulator, and a 5G core network. UERANSIM[3] is used to simulate the UE and gNB, while Open5GS is used as the 5G core network under test. Wireshark is used to capture and analyze NGAP(NG Application Protocol) and NAS(Non Access Stratum) message flows.

In a normal registration procedure, the UE sends an Authentication Response after receiving an Authentication Request. After authentication is completed, the network sends a Security Mode Command, and the UE replies with a Security Mode Complete message.

In contrast, the abnormal UE generated by the proposed tool skips the Authentication Response and sends a Security Mode Complete message first.

The inspection criterion is whether the core network proceeds to Registration Accept or PDU(Protocol Data Unit) Session Establishment after receiving this abnormal NAS message. If these procedures are not performed, the network is considered to properly block the authentication bypass attempt.

3 Evaluation

The experiment was conducted using Open5GS and UERANSIM. The UE simulator was modified at the NAS procedure level so that it generated a Security Mode Complete message without sending a valid Authentication Response. The goal was to observe whether the 5G core network accepts or blocks a UE that attempts to proceed before authentication is completed.

Two cases were tested. In the normal UE case, the UE transmitted Authentication Response after Authentication Request, and the core network continued with Security Mode Command, Security Mode Complete, Registration Accept, and PDU session-related procedures.

In the abnormal UE case, the UE did not send Authentication Response. Instead, it transmitted Security Mode Complete immediately after receiving Authentication Request. Wireshark analysis confirmed that Authentication Response was not observed in the abnormal flow. In addition, repeated Authentication Request and Authentication Failure-related messages were observed.

The results indicate that the Open5GS-based 5G core network did not allow the abnormal UE to complete registration without authentication.

4 Conclusion

This paper proposed a UERANSIM-based inspection tool for verifying authentication bypass attempts in a 5G core network. The proposed tool changes the NAS message processing behavior of the UE simulator so that the UE sends Security Mode Complete without sending Authentication Response.

Experimental results showed that the Open5GS core network did not send Registration Accept or proceed to PDU Session Establishment for the abnormal UE. Instead, authentication failure and re-authentication procedures were observed. Therefore, the tested 5G core network properly blocked access from a UE that had not completed authentication. Future work will extend the tool to other NAS-based security inspection items, such as replay attacks and NAS integrity verification failures. It will also be applied to 5G core testbeds such as free5GC and OpenAir Interface for further validation

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